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American Sign Language Recognition and Translation Using Wearable Sensors

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1. Introduction and Problem Statement

Sign language is the primary method of communication for the deaf and hard-of-hearing community. While previous research has explored sign language recognition, it generally falls into two categories: vision-based and wearable sensors-based methods. Vision-based methods (like cameras) can be affected by lighting, background environment, and privacy concerns. On the other hand, existing wearable sensor datasets are often limited in size and rarely account for the grammatical differences between sign language and spoken language.

This research aims to bridge that gap by using a wearable inertial motion capture system to collect a larger dataset of American Sign Language (ASL) and applying deep learning models to recognize and translate these gestures into English sentences.

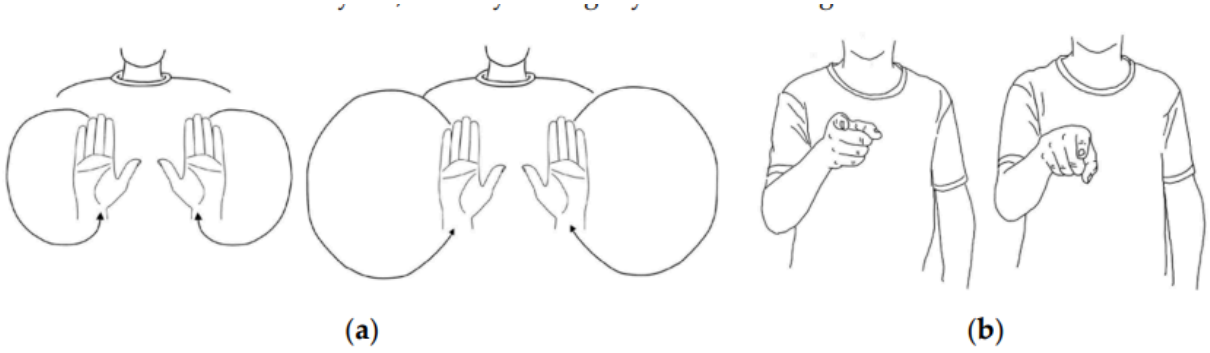


Figure 1. Different expressions for the same word: (**a**) "Sunday"; (**b**) "you".

2. Methodology

The researchers broke their approach down into data collection and the development of two specific neural network models.

Hardware and Data Collection:

- The team used the "Perception Neuron" motion capture system, which consists of 25 inertial measurement units (IMUs) placed on the user's upper body (arms, hands, back, and head).
- They collected a dataset of 300 commonly used ASL sentences performed by 3 volunteers who were ASL beginners.
- These sentences comprised 455 different hand gestures, resulting in a total of 7,400 recorded data samples.
- A 38-dimensional vector was selected to capture the relevant rotation data of the arms and hands.

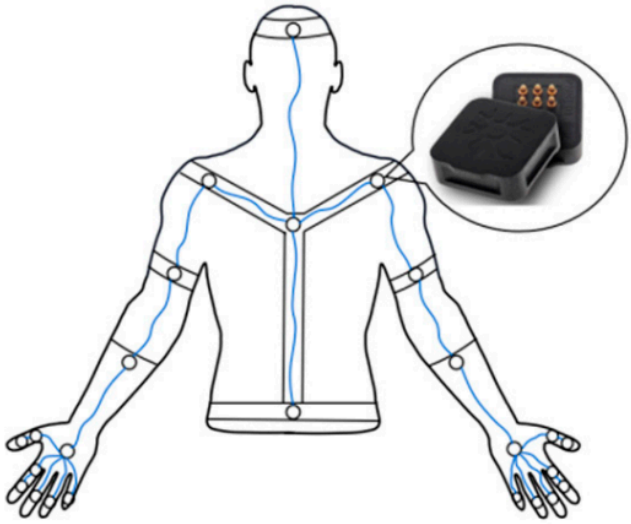


Figure 2. Sensor distribution under the human back view [27].

Deep Learning Models: The researchers created two different labels for the data: one mapping the exact sequence of hand gestures, and another mapping the translation to spoken English grammar. They built two models to handle this:

1. **Sequence Recognition Model:** This model identifies the exact sequence of signed gestures. It uses a Convolutional Neural Network (CNN) to extract features, a Bi-directional Long-Short Term Memory (Bi-LSTM) network to handle the sequence of data over time, and a Connectionist Temporal Classification (CTC) layer to align the

output with the correct text labels.

2. **End-to-End Translation Model:** Since ASL grammar differs from spoken English (for example, missing words like "am" or "is"), this model translates the signs into standard English sentences. It uses an Encoder-Decoder structure with global attention, allowing the system to map the physical inertial data directly to text.

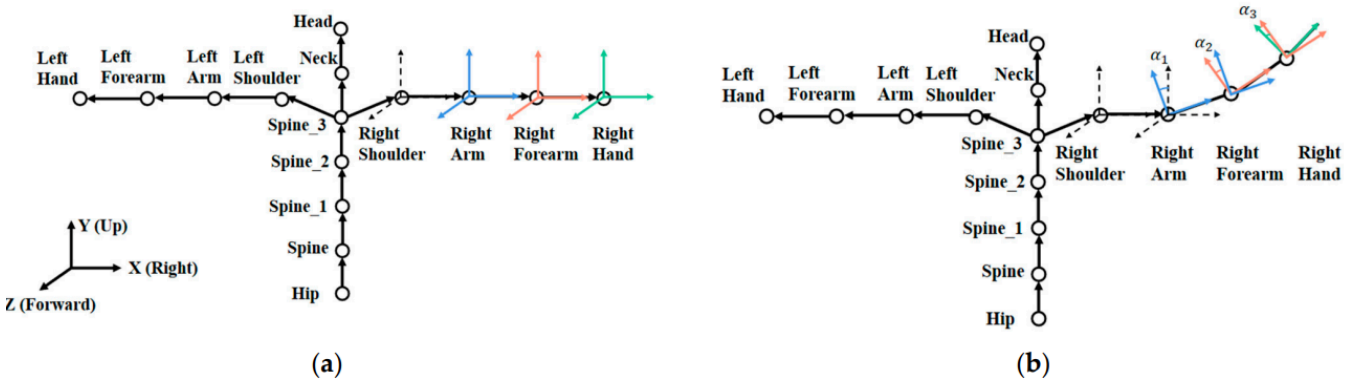


Figure 3. The relative relationship of upper-body joint coordinates: (**a**) initial sensor calibration posture; (**b**) right arm slightly raised.

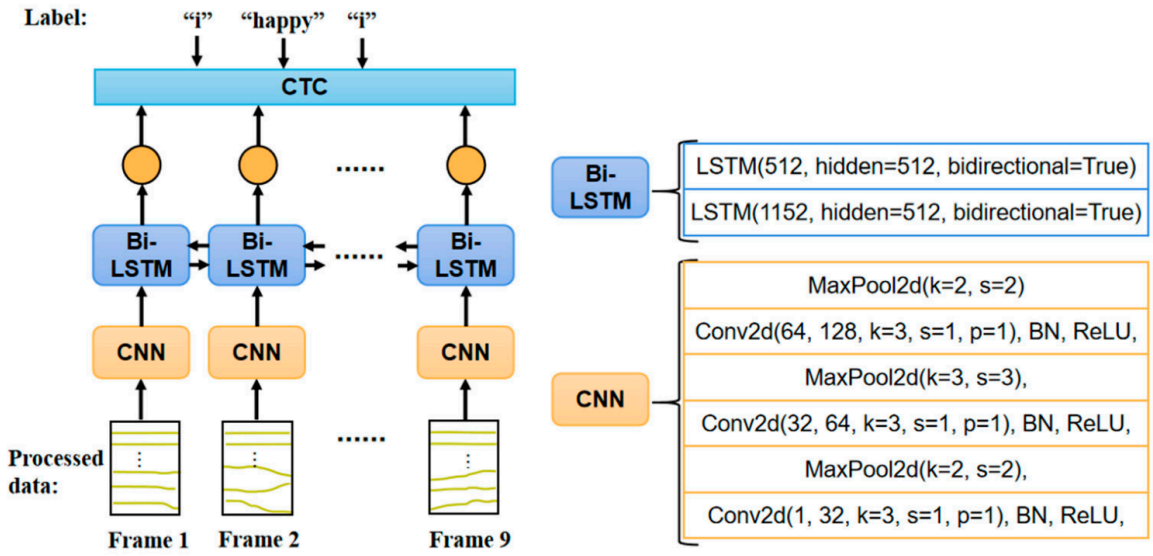
3. Key Findings and Results

The models were tested using a five-fold cross-validation method, and the results showed strong potential.

- **Recognition Accuracy:** The sequence recognition model performed exceptionally well. It achieved an accuracy rate of 99.07% at the word level (only a 0.93% Word Error Rate) and 97.34% at the sentence level. This proves that the wearable sensors accurately captured distinct gestures.

- **Translation Accuracy:** The end-to-end translation task proved to be more difficult. The model had a Word Error Rate of 16.63% and a Sentence Error Rate of 39.27%.

- **Challenges in Translation:** The error rates increased during translation because the model's output sentences were often shorter than the actual labels, and the system struggled with spoken English words that do not have direct physical ASL gestures.



4. Limitations and Future Scope

While the study presents a solid foundation, the authors noted a few limitations:

- **Lack of Grammar Knowledge:** Because the dataset of text sentences was relatively small, the model couldn't learn enough about standard English grammar to perfect the translation. A dedicated language model is needed to make the outputs sound more like natural spoken language.
- **Individual Differences:** When testing the model on a user whose data wasn't heavily featured in the training set, the accuracy dropped significantly due to natural variations in how different people move.
- **Future Work:** The researchers suggest involving more participants in future studies to help the model learn more patterns, as well as utilizing more convenient wearable technologies as they become available.

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